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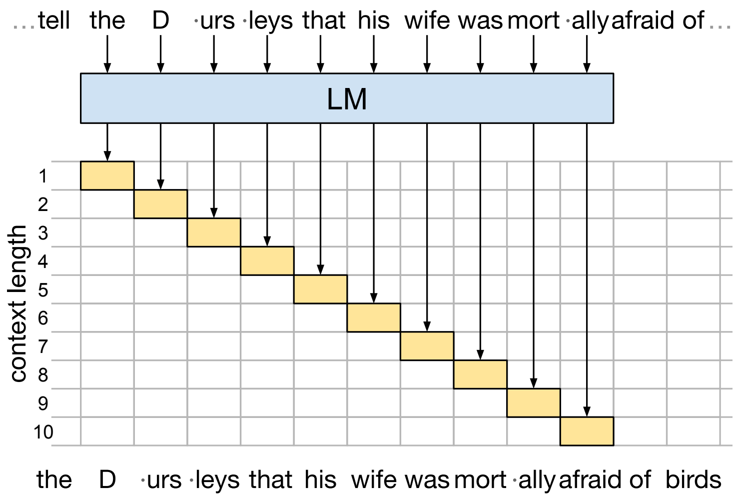

Figure 7: A step of context length probing with $c_{\max} = 10$. The input tokens are shown at the top, the target tokens at the bottom. When the LM is run on a segment of the document, the effective context length for each target token is equal to its offset from the beginning of the segment, e.g. the context for predicting "␣D" is "␣the" ($c = 1$), the context for "urs" is "␣the␣D" ($c = 2$), etc.

A Context length probing

Fig. 7 illustrates a step of context length probing. We wish to obtain the tensor $\boldsymbol{P}$ from Eq. (3), understood as a table where each cell contains the predictions (next-token logits) for a given position in the text and a given context length. By running our LM on a segment of the text, we get predictions such that for the n-th token in the segment, the effective context length is equal to n, which corresponds to a diagonal in the table. We can thus fill in the whole table by running the LM on all segments of length $c_{\max}$ (plus trailing segments of lengths $c_{\max} - 1, \ldots, 1$).

Notice that this process is somewhat similar to (naïvely) running the LM in generation mode, except that at each step, the leading token is removed, preventing the use of caching to speed up the computation.

In practice, it is not necessary to explicitly construct the tensor $\boldsymbol{P}$. Indeed, we find it more efficient to instead store the raw logits obtained by running the model on all the segments, then do the necessary index arithmetics when computing the metrics.

A.1 Strided context length probing

For a k-fold reduction in computational cost, we may instead use a sliding window with a stride $k > 1$, i.e. run the model only on segments starting at positions $k\,(n-1)+1$ for all $n \in \{1, \ldots, \lceil N/k \rceil\}$, rather than all positions. This way, for a target token x_{n+1}, we obtain the predictions $p(x_{n+1} \mid x_{n-c+1}, \ldots, x_n)$ only for such context lengths c that $c \bmod k = n$. In other words, predictions with context length c are only available for tokens $x_{c+1}, x_{c+k+1}, x_{c+2k+1}, \ldots$. Consequently:

- Overall, we still cover all context lengths $1, \ldots, c_{\max}$, allowing us to perform aggregate analyses like the ones in Section 4.1.
- When analyzing the predictions for a specific target token in a document (e.g. to compute Δ-scores), context tokens come in blocks of length k. Visualizations like the ones in Figs. 1 and 4 are still possible for all target tokens, but become less detailed, grouping every k context tokens together.
- Computation time, as well as the space needed to store the predictions, is reduced by a factor of k.

B Technical details

Data. The LinES treebank is licensed under Creative Commons BY-NC-SA 4.0. We concatenated all tokens from each of the documents from the treebank, then re-tokenized them using the GPT-2 tokenizer.

We mapped the original (UD) POS tags to the GPT-tokenized dataset in such a way that every GPT token is assigned the POS tag of the first UD token it overlaps with.

Models. We used the models `EleutherAI/gpt-j-6B` (Apache 2.0 license), and `gpt2-xl` and `gpt2` (MIT license), all from `huggingface.co`.

Computation. We parallelized the inference over 500 jobs on a compute cluster,[8] each running on 8 CPU cores with at least $8\,\mathrm{GB}$ of RAM per core, with a batch size of 16. Each job took about 10–$20\,\mathrm{min}$ for GPT-2 and 30–$60\,\mathrm{min}$ for GPT-J. Additionally, computing the metrics from the logits (which take up $2\,\mathrm{TB}$ of disk space in `float16`) took between 2 and $4\,\mathrm{h}$ per model on a single machine with 32 CPU cores. The total computing time was 318 core-days, including debugging and discarded runs.

C Additional plots

C.1 Token-wise metrics as a function of context length

Figs. 8 and 9 show NLL and KL divergence (5), respectively, as a function of context length, for selected target tokens (proper nouns) from the validation set.

[8] Nef, the cluster computing infrastructure of Inria Sophia Antipolis Méditerranée; see `https://wiki.inria.fr/ClustersSophia`